

Certain low confidence words experiment by GPT-3.5-Turbo

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Idea

In this step, Enno suggested me to ask ChatGPT to only fix certain low-confidence words in the ASR output rather than the whole sentence.

For this experiment, I use GPT-3.5-Turbo 1106 as a language model to correct the ASR errors.

Here is the prompt of this experiment

```
{  'role': 'system',  
  'content': f"""You are a helpful assistant that corrects ASR errors. \  
  
You will be presented with an ASR transcription and low confidence words in that transcription. \  
  
the input will be formatted as json with keys text and low_confidence_words,\  
  
where the text is the ASR transcription and low_confidence_words contains the list of words with low asr confidence. \  
  
your task is to check if the low confidence words make sense in the transcription and if they do not, replace them with better words. \  
  
provide your output as a string. \  
  
Do not change the case, for example, lower case or upper case, in the transcription. \  
  
Do not output any additional text that is not the corrected transcription. \  
  
Do not write any explanatory text that is not the corrected transcription.  
"""}

```

Providing example in prompt

```
{  
    'role': 'user',  
    'content': '{"text": "Why not allow your silver tuff to luxuriate in a natural manner?",  
"low_confidence_words":["tuff"]}'  
},  
{'role': 'assistant', 'content': "why not allow your silver tufts to luxuriate in a natural manner?"},  
{'role': 'user', 'content': json.dumps(asr_transcription)}  
}
```

Code snippet

```
for i,d in enumerate(data):
```

```
    asr_transcription = confidence_score_word_level(d["asr_transcription"], confidence = True)
```

```
    low_confidence_words = [word["text"] for word in asr_transcription["words"]
```

```
        if word["confidence"] <= THRESH]
```

```
    if len(low_confidence_words) > 0:
```

```
        asr_transcription = {"text": asr_transcription["text"], "low_confidence_words": low_confidence_words}
```

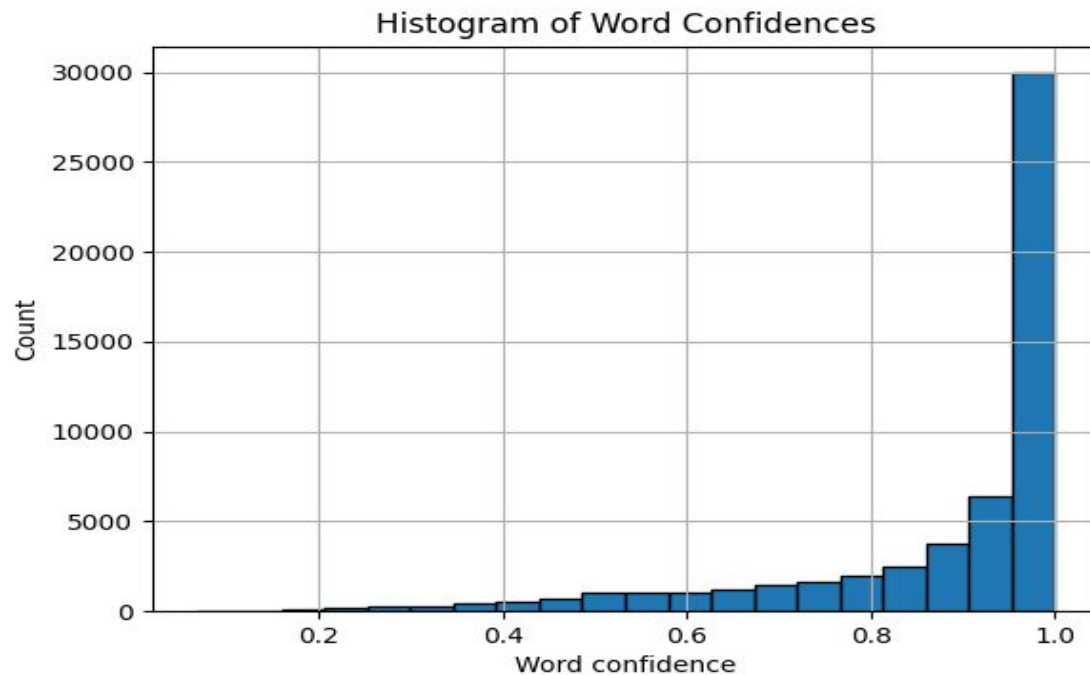
```
        reference_transcription= d["reference_transcription"]
```

```
        data1.append({"asr_transcription": asr_transcription, "reference_transcription": reference_transcription})
```

Here is the one example of output for this experiment

```
{  
  "asr_transcription": {  
    "text": " But, Dukalion and Piro were very sad. For they knew that they were the only persons who were left alive in all the land.",  
    "low_confidence_words": [  
      "Dukalion",  
      "Piro"  
    ]  
  },  
  "reference_transcription": "BUT DEUCALION AND PYRRHA WERE VERY SAD FOR THEY KNEW THAT THEY WERE THE ONLY PERSONS WHO WERE LEFT ALIVE IN ALL THE LAND",  
  "corrected_asr_transcription": "But, Deucalion and Pyrrha were very sad. For they knew that they were the only persons who were left alive in all the land."  
}
```

Histogram of word confidence



Results for this experiment by GPT-3.5 Turbo (Thresh =0.5)

```
WER_original:      8.6116%  
WER_corrected_chatgpt: 7.5211%  
CER_original:      3.5707%  
CER_corrected_chatgpt: 3.6174%  
SER_original:      61.4502%  
SER_corrected_chatgpt: 59.1565%  
---
```


Results for this experiment by GPT-3.5 Turbo (Thresh =0.55)

```
WER_original:          10.2260%  
WER_corrected_chatgpt:  9.1994%  
CER_original:          4.2068%  
CER_corrected_chatgpt:  4.4297%  
SER_original:          77.3326%  
SER_corrected_chatgpt:  77.4973%  
---
```

Results for this experiment by GPT-3.5 Turbo (Thresh =0.6)

```
WER_original:          9.7043%  
WER_corrected_chatgpt: 8.9645%  
CER_original:          3.9945%  
CER_corrected_chatgpt: 4.3996%  
SER_original:          73.6453%  
SER_corrected_chatgpt: 75.4680%  
---
```

Results for this experiment by GPT-3.5 Turbo (Thresh =0.65)

```
WER_original:          9.3484%  
WER_corrected_chatgpt: 8.7986%  
CER_original:          3.8361%  
CER_corrected_chatgpt: 4.3280%  
SER_original:          70.8711%  
SER_corrected_chatgpt: 73.6388%  
---
```

Results for this experiment by GPT-3.5 Turbo (Thresh =0.7)

```
WER_original:          9.0700%  
WER_corrected_chatgpt: 8.7170%  
CER_original:          3.7202%  
CER_corrected_chatgpt: 4.2891%  
SER_original:          68.1992%  
SER_corrected_chatgpt: 71.6901%  
---
```

Results for this experiment by GPT-3.5 Turbo (Thresh =0.75)

```
WER_original:      8.9594%  
WER_corrected_chatgpt: 8.6679%  
CER_original:      3.6863%  
CER_corrected_chatgpt: 4.3108%  
SER_original:      66.7897%  
SER_corrected_chatgpt: 70.6847%  
- - -
```

Results for this experiment by GPT-3.5 Turbo (Thresh =0.8)

```
WER_original:      8.8127%  
WER_corrected_chatgpt: 8.5871%  
CER_original:      3.6396%  
CER_corrected_chatgpt: 4.2491%  
SER_original:      64.5972%  
SER_corrected_chatgpt: 69.7053%  
---
```

Results for this experiment by GPT-3.5 Turbo (Thresh =0.85)

```
WER_original:      8.7179%  
WER_corrected_chatgpt: 8.4257%  
CER_original:      3.6077%  
CER_corrected_chatgpt: 4.1802%  
SER_original:      63.2747%  
SER_corrected_chatgpt: 67.4063%  
---
```

Results for this experiment by GPT-3.5 Turbo (Thresh =0.9)

```
WER_original:      8.6424%  
WER_corrected_chatgpt: 8.4737%  
CER_original:      3.5841%  
CER_corrected_chatgpt: 4.2055%  
SER_original:      61.9724%  
SER_corrected_chatgpt: 65.5585%  
---
```


Results for this experiment by GPT-3.5 Turbo (Thresh =0.95)

```
WER_original:      8.6157%  
WER_corrected_chatgpt: 8.5041%  
CER_original:      3.5723%  
CER_corrected_chatgpt: 4.1458%  
SER_original:      61.5185%  
SER_corrected_chatgpt: 65.0370%  
---
```

The results were not very good for GPT-3.5 Turbo.

Now, I've decided to conduct this experiment using GPT-4 Turbo with a threshold of 0.6.

Results for this experiment by GPT-4 Turbo (Thresh =0.6)

```
WER_original:          9.7043%  
WER_corrected_chatgpt: 7.6133%  
CER_original:          3.9945%  
CER_corrected_chatgpt: 3.8408%  
SER_original:          73.6453%  
SER_corrected_chatgpt: 65.9113%  
---
```

It seems that for this task, GPT-4 Turbo has the better performance rather than GPT-3.5-Turbo.

But to prove this, more experiment should be done.